

RENANCE

OSUN STATE UNIVERSITY, OSOGBO

MCB 201 — Past Questions & Materials

by Renance

Course: MCB 201

Institution: Osun State University, Osogbo

Papers in this pack: 1

Levels: 200

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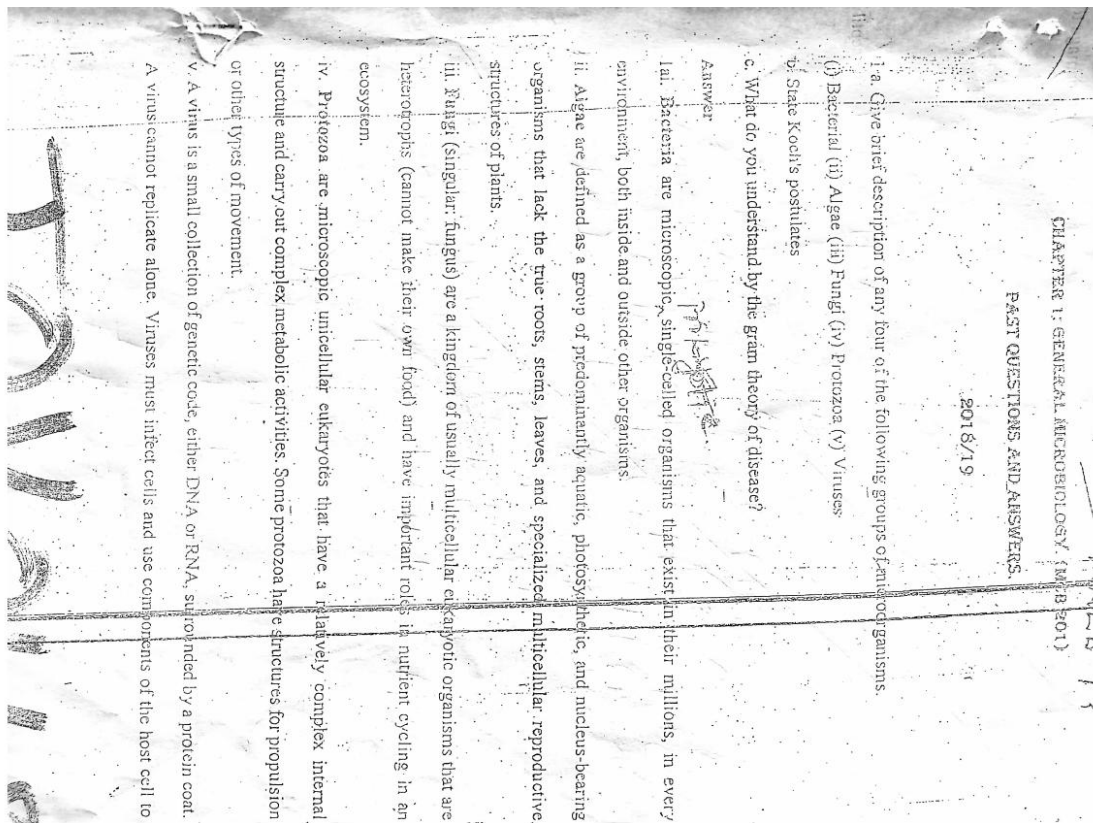
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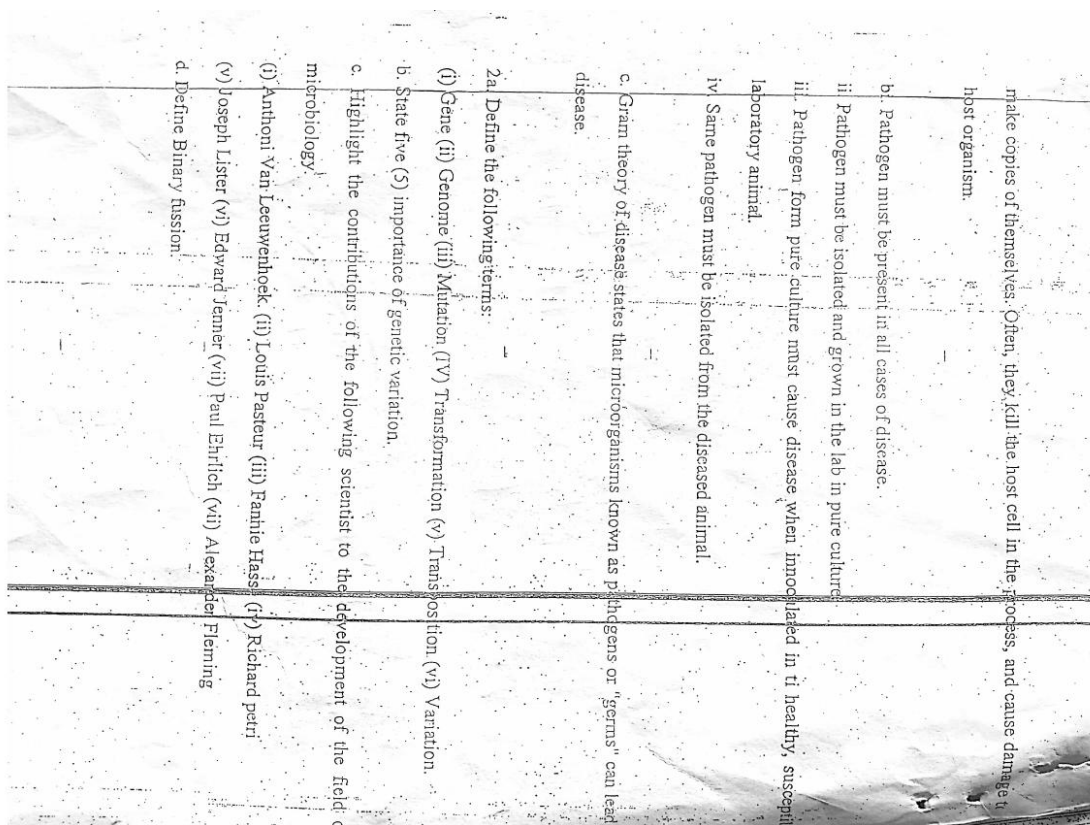
Paper 1 of 1 — MCB 201 PQ & ANS

Level 200 • First Semester • PastQuestion

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Answer:

2a. Gene a unit of heredity which is transferred from a parent to offspring and is held to determine some characteristic of the offspring.

ii. Genome is the haploid set of chromosomes in a gamete or microorganism, or in each cell of a multicellular organism.

iii. Mutation is the changing of the structure of a gene, resulting in a variant form that may be transmitted to subsequent generations, caused by the alteration of single base units in DNA, or the deletion, insertion, or rearrangement of larger sections of genes or chromosomes.

iv. Transformation is one of several processes by which genetic material in the form of "naked" deoxyribonucleic acid (DNA) is transferred between microbial cells.

v. Transposition is a process by which segments of a chromosome are relocated through the genome.

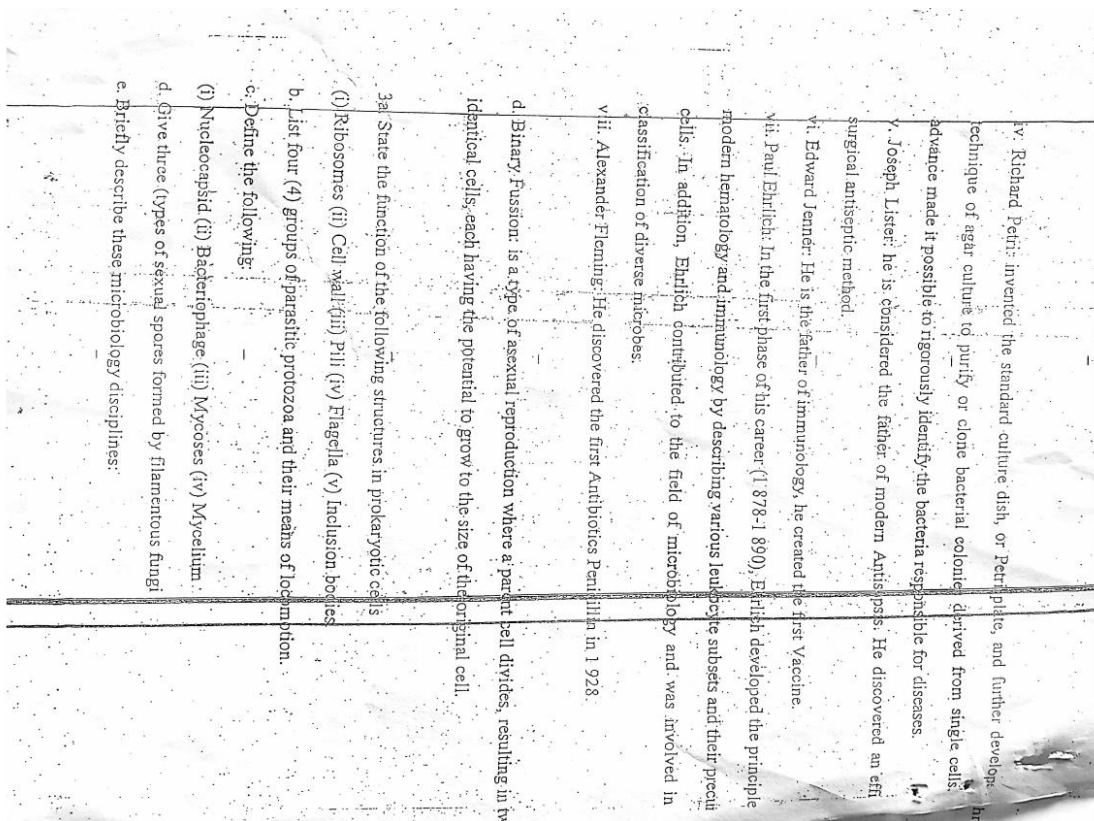
vi. Variation is a process by which segments of a chromosome are relocated through the genome.

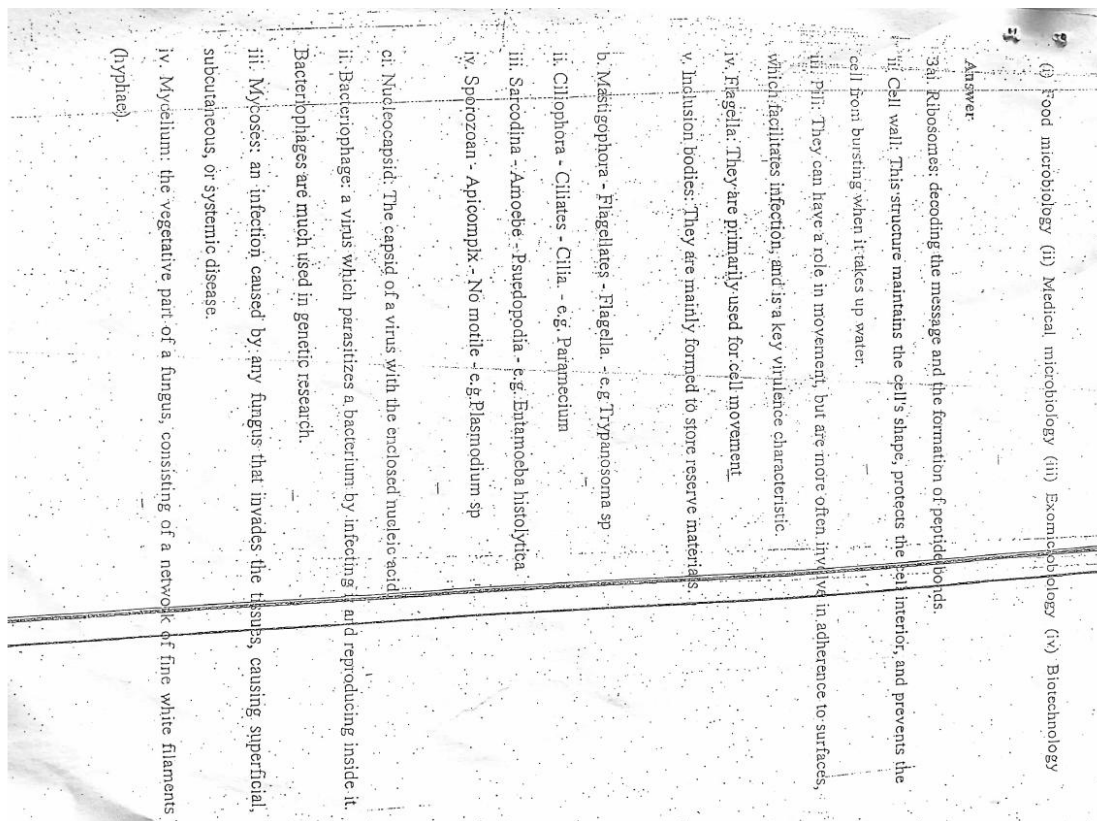
c. Antoni Van Leeuwenhoek: was one of the first people to observe microorganisms, using a microscope of his own design, and made one of the most important contributions to biology.

ii. Louis Pasteur: discovered that microorganisms cause fermentation and disease; originated the process of pasteurization; saved the beer, wine, and silk industries in France; and developed vaccines against anthrax and rabies.

iii. Fannie Hesse: Contributed Agar and Art to Microbiology. Hesse not only contributed agar to microbiology, but she also worked in Koch's lab as an assistant, much like one of today's lab technicians, and was a scientific illustrator.

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- di. Sporangiospores
- ii. Conidiospores
- iii. Chlamydospores

ei. Food microbiology: Is the branch of microbiology that deals with methods for key microorganisms (especially food-borne pathogens and spoilage microbes) from growing in food during handling, processing and storage.

ii. Medical microbiology: is a branch of medical science concerned with the prevention, diagnosis and treatment of infectious diseases. In addition, this field of science studies various clinical applications of microbes for the improvement of health.

iii. Exomicrobiology: Is the study of microorganisms in outer space. Microorganisms in outer space are most wide spread form of life on Earth, and are capable of colonizing any environment.

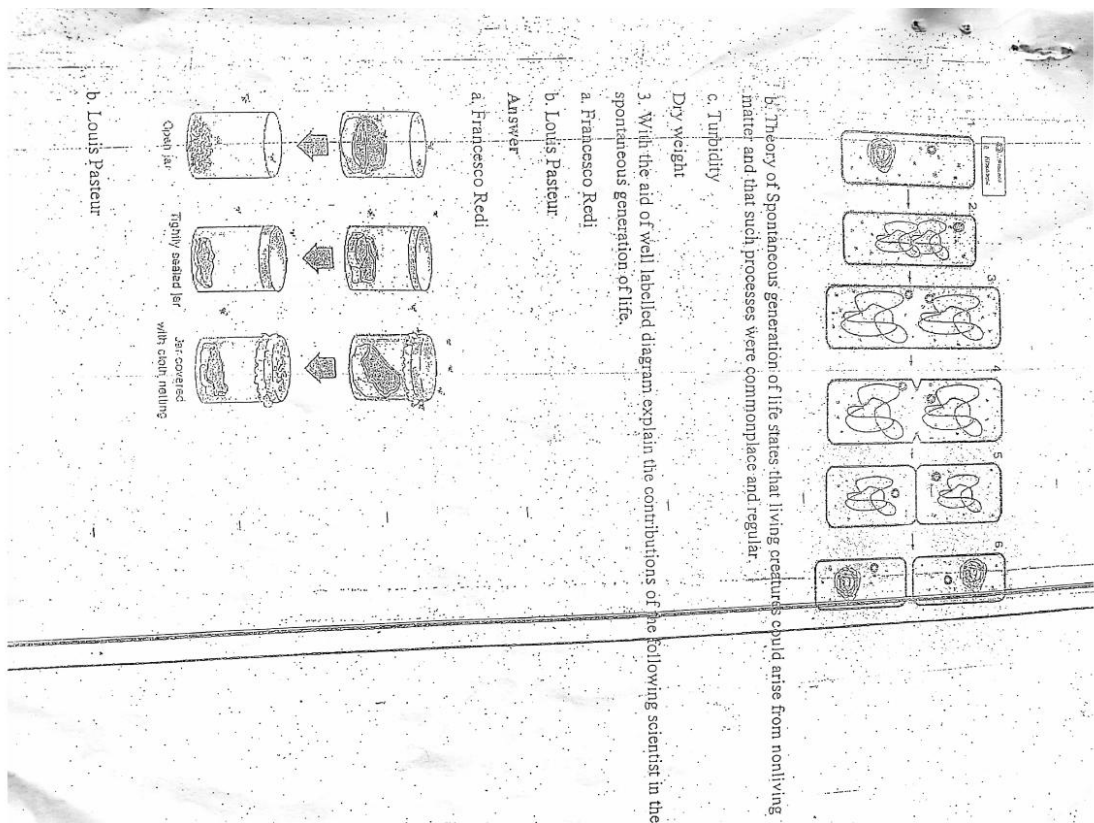
iv. Biotechnology: Is technology that utilizes biological systems, living organisms (Unusually microorganisms) or parts of this to develop or create different products.

Section B

- 4 a. Briefly explain the nutritional classes of microorganisms.
- b. State the principles upon which microbial metabolism can be summarized
- c. Highlight the methods widely employed in the determination of microbial growth



i. The microorganism must be found in abundance in all organisms suffering from the disease, but should not be found in healthy organisms.	
ii. The microorganism must be isolated from a diseased organism and grown in pure culture.	
iii. The cultured microorganism should cause disease when introduced into a healthy organism.	
iv. The microorganism must be reisolated from the inoculated, diseased experimental host and identified as being identical to the original specific causative agent.	
c. Yeast	
Mould	
Dimorphic Fungi	
d. Microscope	
Inoculating loop	
Inoculating chamber	
Test tube	
Refrigerator	
Petri dish	
ei. Wuhan, China	
ii. Aimeyo Stella Adedevon	
iii. Patrick Oliver Sawyer	
f. Bacteria variation: a change in the genotype of a bacterium or its phenotype	
g. Probiotics helps balance the friendly bacteria in the body	
ii. Probiotics can help prevent and treat diarrhea	
iii. Probiotics supplements improves some mental health conditions	
iv. Probiotics can help boost the immune system	
v. Certain probiotics strain can help keep the heart healthy	
2. Using well labelled diagram, describe the process of binary fission in a typical bacteria cell.	
b. State the hypothesis of Aristotle in the Theory of spontaneous generation of life	
c. Elucidate the two indirect methods of measurement of cell mass	
Answer	



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4. Differentiate between Heritable variation and Non-heritable variation

Answer

Heritable variation: this are the traits that can be passed from one generation to another

While

Non-heritable Variation: are usually associated with a learned behaviour

b. To understand gene function of microorganisms

ii. Microorganisms are used for the isolation and multiplication of some gene of interest of some higher organism.

iii. They help in understanding the Genetics of disease causing microorganisms especially Viruses, will be useful to control disease

iv. They provide relatively simple system for studying genetics phenomenon and thus useful to other organisms.

5. Define the terms

I. Genotype

iii. Phenotype

b. State 3 importance of genetic variation in a population.

Answer

5. a.i Genotype: this is an individual collection of gene.

Phenotype: the set of observable characteristics of an individual resulting from the environment

b. It helps in adaptation of some organism

PAST QUESTION AND ANSWER

2021/2022

INSTRUCTION: ANSWER QUESTION ONE AND THREE OTHER QUESTION

1.(a). What are plasmids?

ii. Give two examples of plasmids.

iii. Who discovered the first antibiotics?

iv. In what year was it discovered?

(b). In a tabular form, state two differences in the structural composition of gram positive and Gram negative bacteria.

ii. List three structures external to a bacteria cell.

iii. State four the economic importance of algae.

(c). i. Microorganism belong to three of the five kingdom of living things. Mention the three kingdoms.

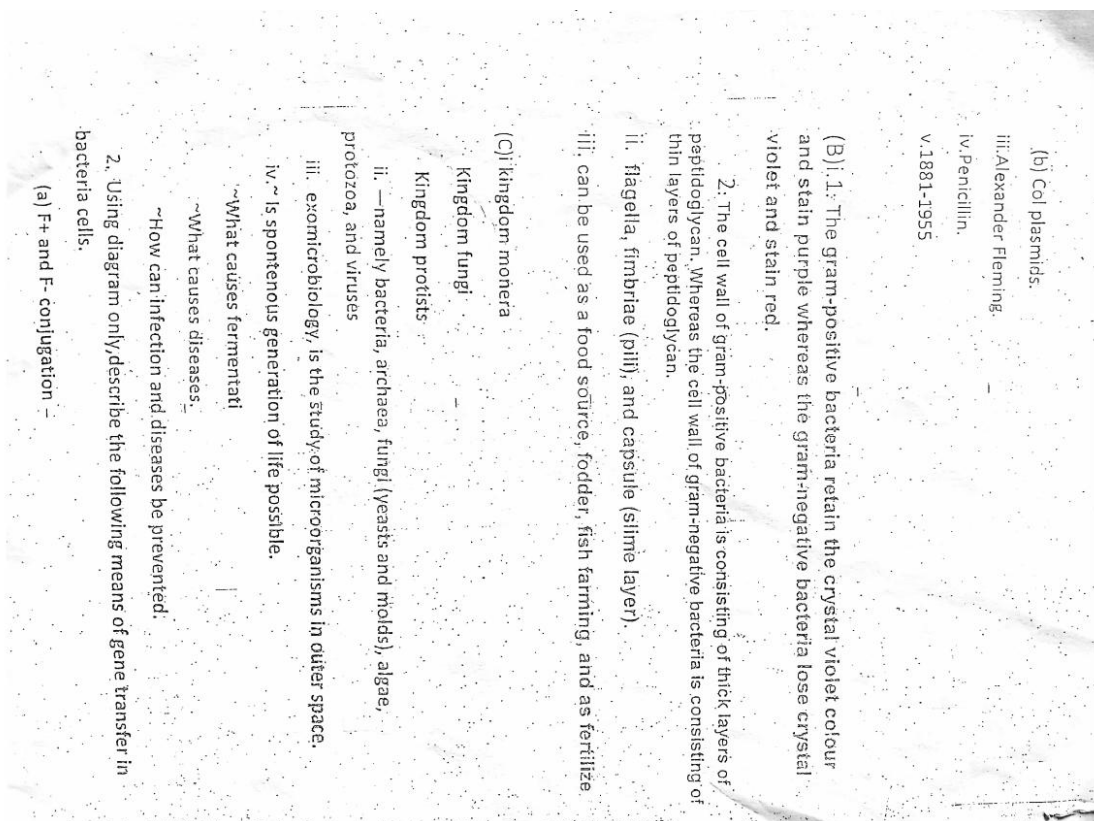
ii. What are the six major groups of microorganism?

iii. Explain the term Exomicrobiology.

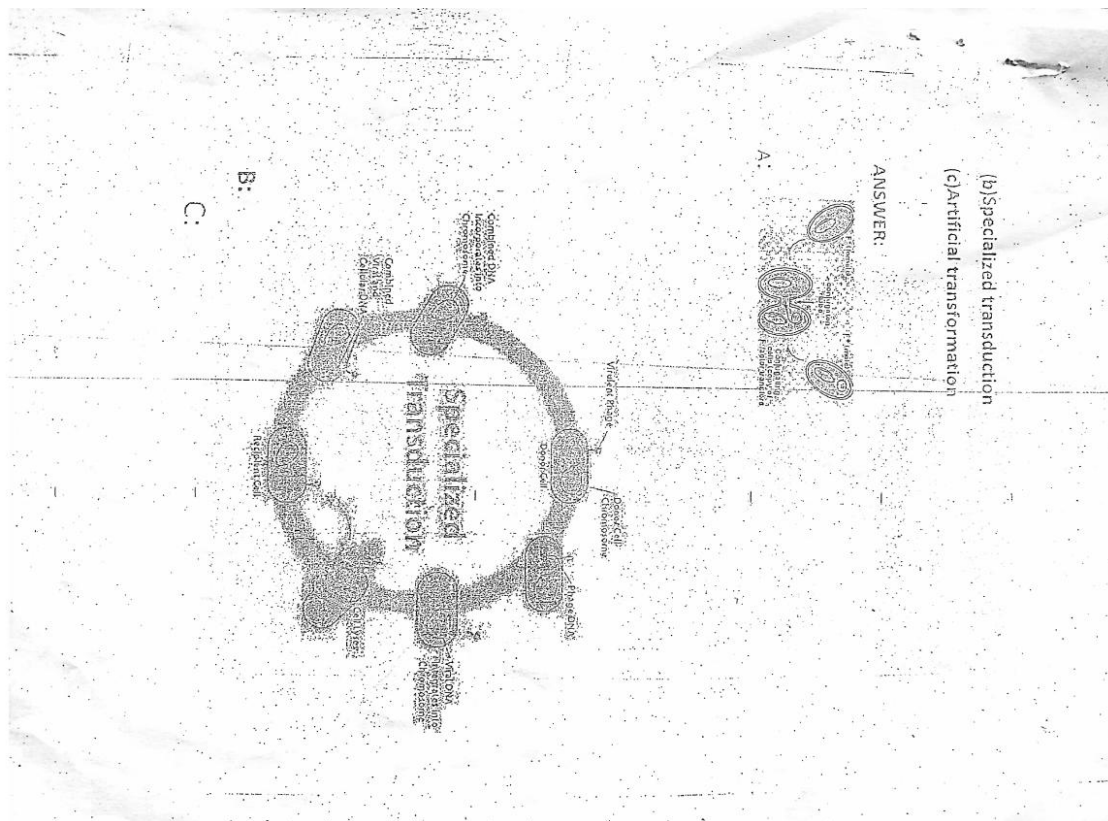
iv. In the Golden age of microbiology, state the four questions scientists were seeking to answer.

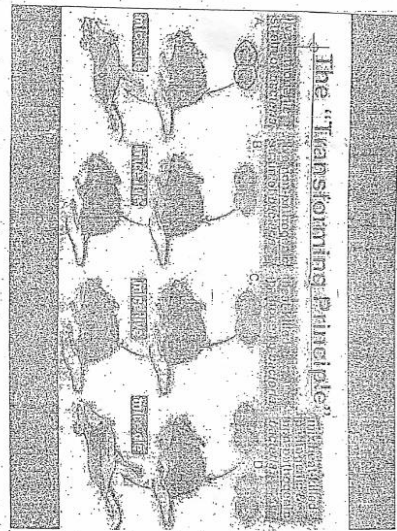
ANSWER:

- 1.(a) i. A genetic structure in a cell that can replicate independently of the chromosomes. Typically a small circular DNA strand in the cytoplasm of a bacterium or protozoan. Plasmids are much used in the laboratory manipulation of genes
- ii. (a) fertility plasmids



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3. Using table, classify microorganism based on their nutritional requirement stating their carbon, electron, and energy sources give an example of microorganism in each class

ANSWER

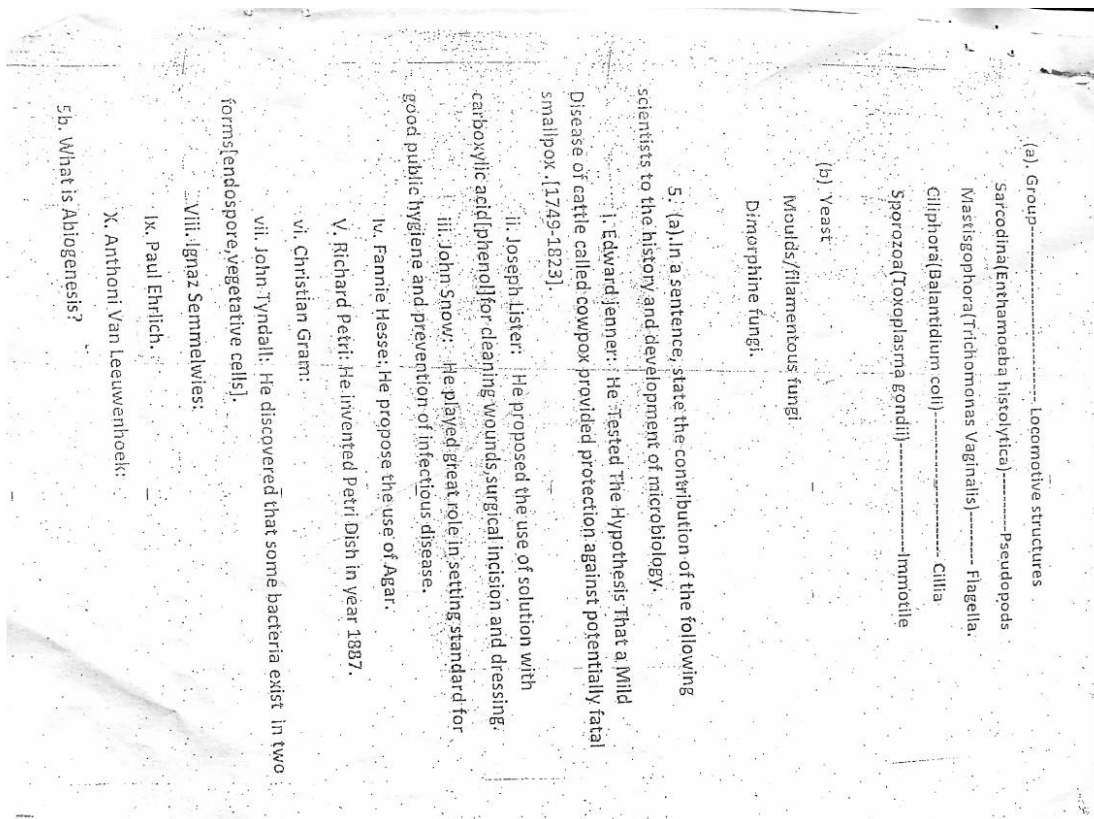
Table 2. Major nutritional types of bacteria based on carbon and energy sources

S. No.	Nutritional type	Energy source	Carbon source	Examples
1.	Photoautotrophs	Sunlight	CO ₂ fixation	Cyanobacteria, some chloroflexi, purple and green sulphur bacteria
2.	Photoheterotrophs	Light	Organic compounds	Some sulphur bacteria, some cyanobacteria
3.	Chemoautotrophs	Inorganic compounds (e.g. H ₂ , NH ₃ , NO ₂ , H ₂ S, Fe ²⁺)	CO ₂	A few bacteria and many archaea
4.	Chemoheterotrophs	Organic compounds	Organic compounds	Most bacteria, some archaea, eukaryotes

4. (a) Briefly classify protozoa into four groups based on their locomotive structure with relevant example of each group.

(b) What are the three morphological classification of fungi.

ANSWER:



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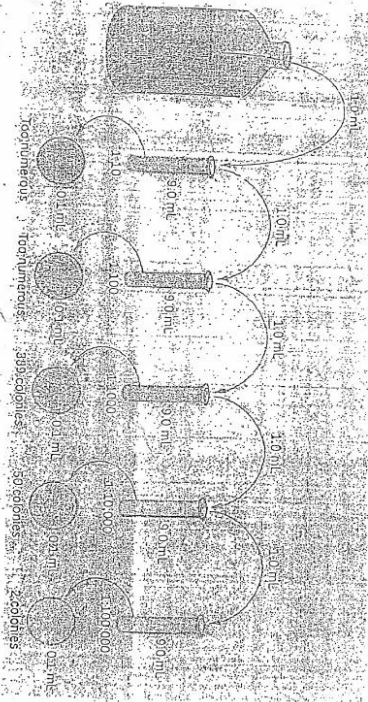
Ans] This is the Idea that life routinely appear or arises from non life.

5c. Mention two scientist who proposed and two scientist who disproved the theory of spontaneous generation of life..

Ans1.Francesco Redi & Turberville Needham and Louis Pasteur & Lazzaro Spallanzani

6[a] Explain with Appropriate Diagram the direct measurement of cell numbers using serial dilution/spread plate or pour plate technique.

Ans]



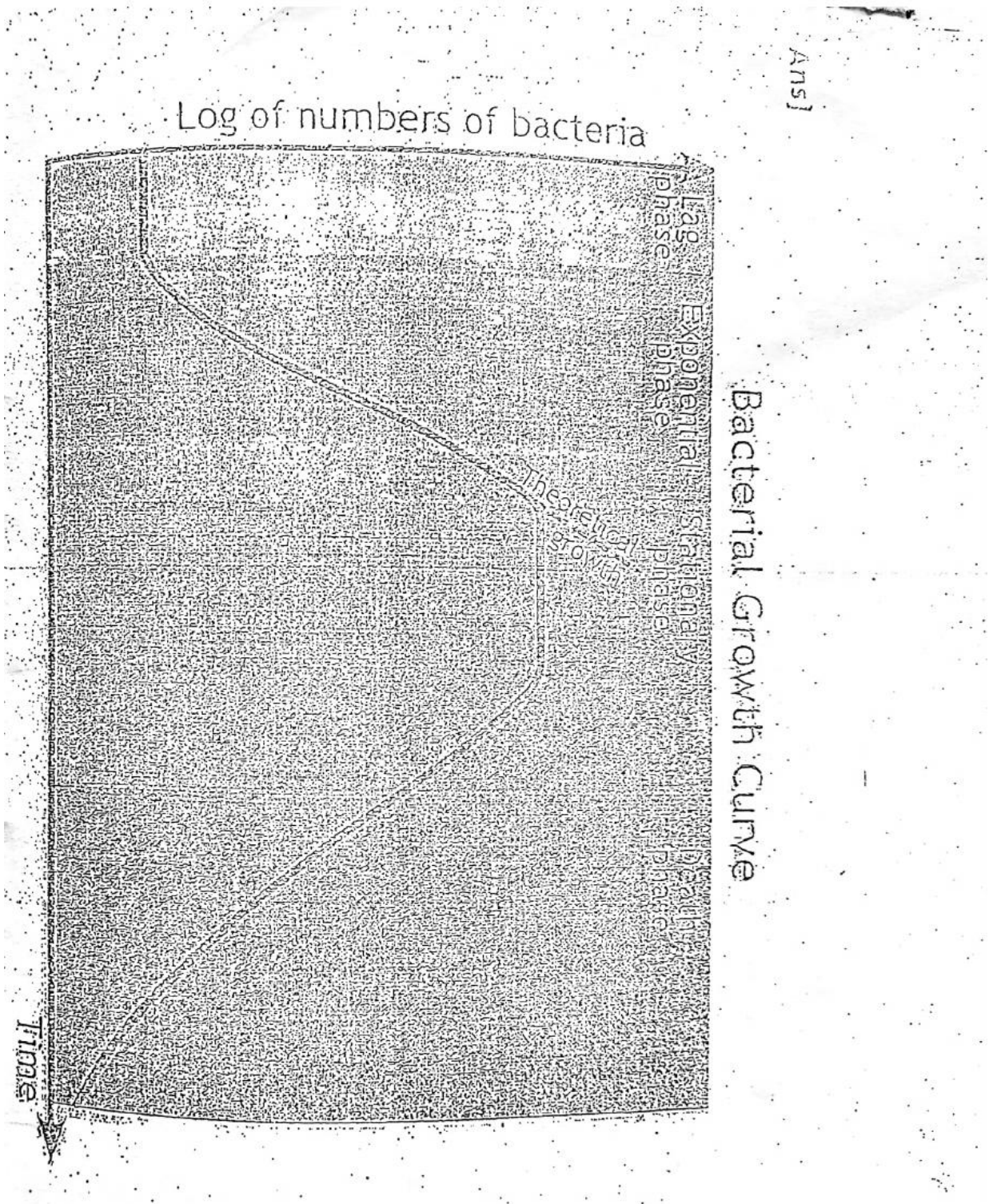
6b) Mention the intracellular enzyme for each of the following laboratory test: i) Indole production test : Intracellular enzyme tryptophanase

- ii] Voges Prokauer :

iii] Oxidase test : cytochrome oxidase enzymes:

[Ans]

6c] Illustrate a typical microbial growth curve using a well-labeled diagram.





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